ARTICLE

Translational Therapeutics

Remodelling hypoxic TNBC microenvironment restores antitumor efficacy of Vγ9Vδ2 T cell therapy

Yanyun Jing[1,6], Yangzhe Wu[1,6✉], Qinglin Hu[2], Wenfeng Wu[3], Dang Li[4], Wenhao Hu[4], Heming Li[1], Minggang Fu[4✉], Xin Huang[5✉] and Yi Hu[2✉]

BACKGROUND: γδ T cells have emerged as pivotal regulators within the breast cancer tumour microenvironment (TME) and represent a promising therapeutic strategy for late-stage and metastatic breast cancer. In recent years, our research has focused on leveraging allogeneic Vγ9Vδ2 T cells as a novel approach to treat advanced cancers, including triple-negative breast cancer (TNBC). However, the varying clinical outcomes of this therapy have prompted us to investigate the diverse roles of γδ T cells within the TNBC microenvironment and to explore strategies for enhancing therapeutic efficacy through microenvironmental remodelling.
METHODS: Data from TCGA, publicly available scRNA-seq datasets and a series of experiments including flow cytometry, atomic force microscopy, confocal laser scanning microscopy, mouse model and others were applied to examine the functional heterogeneity of γδ T cells in TNBC, non-TNBC, and healthy individuals.
RESULTS: γδ T cells serve as predictive markers of better prognosis in breast cancer. In TNBC tumours, γδ T cells exhibited heightened expression of genes linked to both effector and inhibitory molecules, alongside a significant upregulation of glycolytic activity—patterns not observed in non-TNBC or normal breast tissues. Further analysis demonstrated that hypoxic conditions in the TNBC microenvironment likely contribute to these metabolic changes, leading to upregulation of inhibitory checkpoints and downregulation of effector functions in γδ T cells. Importantly, we showed that suppressing HIF-1 signalling using PX478 enhanced the antitumor efficacy of $V\delta2^{+}\gamma\delta$ T cell therapy in TNBC-bearing mice.
DISCUSSION: This work underscores that remodelling the hypoxic TNBC microenvironment can restore the antitumor activity of Vγ9Vδ2 T cell therapy. Our findings offer a compelling new adjuvant strategy to improve the outcomes of Vγ9Vδ2 T cell-based therapies for advanced breast cancer treatment.

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INTRODUCTION

Triple-negative breast cancer presents a significant treatment challenge

Breast cancer remains the predominant malignancy among women worldwide, retaining its highest mortality rate despite considerable advancements in treatment, as evidenced by data from GLOBOCAN 2022 estimates [1]. Subtyping of breast cancer commonly revolves around the presence or absence of three key receptors: the oestrogen receptor (ER), the progesterone receptor (PR) and the human epidermal growth factor receptor 2 (HER2). Consequently, the principal subtypes include Luminal A (ER^{+} and/or PR^{+}, $HER2^{-}$), Luminal B (ER^{+} and/or PR^{+}, $HER2^{+}$), HER2-enriched (ER^{-}, PR^{-}, $HER2^{+}$), and Triple-negative (ER^{-}, PR^{-}, $HER2^{-}$) [2]. Triple-negative breast cancer (TNBC) is distinguished by its aggressive nature, heightened propensity for metastatic dissemination and an overall dismal prognosis attributed to the absence of targeted therapeutic options. Alarmingly, 70% of individuals diagnosed with TNBC face mortality within 5 years, presenting a stark contrast to the 44% mortality rate observed in other subgroups [3]. Accounting for 10–15% of all breast cancers in humans [4], the quest for efficacious treatments tailored specifically to address TNBC has emerged as a paramount focus for researchers in the field.

Allogeneic γδ T cell therapy represents a new direction for TNBC immunotherapy

γδ T cells, distinct from αβ T cells, bridge innate and adaptive immunity by recognising antigens independently of MHC

[1]Guangdong Provincial Key Laboratory of Tumour Interventional Diagnosis and Treatment, Zhuhai Institute of Translational Medicine, Zhuhai People's Hospital (The Affiliated Hospital of Beijing Institute of Technology, Zhuhai Clinical Medical College of Jinan University), Zhuhai, Guangdong, PR China. [2]Microbiology and Immunology Department, School of Medicine, Faculty of Medical Science, Jinan University, Guangzhou, Guangdong, PR China. [3]The Institute of Health, Guangdong Pharmaceutical University, Guangzhou, Guangdong, PR China. [4]The Breast Surgery Department, Zhuhai People's Hospital (The Affiliated Hospital of Beijing Institute of Technology, Zhuhai Clinical Medical College of Jinan University), Zhuhai, Guangdong, PR China. [5]The Breast Surgery Department, The First Affiliated Hospital, Faculty of Medical Science, Jinan University, Guangzhou, Guangdong, PR China. [6]These authors contributed equally: Yanyun Jing, Yangzhe Wu. ✉email: tyzwu@jnu.edu.cn; xjfuminggang@126.com; huangxin00@jnu.edu.cn; yihu2020@jnu.edu.cn

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molecules. They also function as antigen-presenting cells, offering therapeutic potential in cancer treatments like allogeneic cell infusion [5, 6]. Human γδ T cells are categorised into Vδ1$^+$, Vδ2$^+$ and Vδ3$^+$ subsets. Vδ2$^+$ cells, predominantly found in peripheral blood, pair with Vγ9 TCR, while Vδ1$^+$ and Vδ3$^+$ subsets mainly reside in tissues [7]. γδ T cells are early producers of IFNγ in tumour immunity [8] and exert antitumor effects through cytotoxic molecules such as granzymes, perforin, CD107a and pathways like Fas/FasL, TRAIL, TNFα and ADCC [7]. Notably, tumour-infiltrating γδ T cells have been linked to favourable prognosis in breast cancer [9], underscoring their clinical relevance.

Taking advantage of the unique properties of γδ T cells, our recent research has pioneered the use of allogeneic Vγ9Vδ2 (also known as Vδ2) T cells for late-stage solid tumours, including breast cancers, thereby validating the safety and efficacy of these cells in tumour therapy [10, 11]. Yet, there remains untapped potential and room for optimisation to fully exploit the effector capabilities of γδ T cells in immunotherapy. Particularly, their precise role in orchestrating immune responses within the breast cancer tumour microenvironment (TME) remains controversial. Both pro- and anti-tumour characteristics of tumour-infiltrating γδ T cells have been documented [12]. For example, cytolytic and IFN-γ producing γδ tumour-infiltrating lymphocytes (TILs) correlate with favourable outcomes in TNBC patients [13, 14], while IL-17 expressing γδ T cells collaborate with neutrophils to promote metastasis and immunosuppression in breast cancer [15, 16]. Furthermore, regulatory functions of γδ TILs have been suggested, indicating suppression of dendritic cell, naïve, and effector T cell responses in the TME of breast cancers [17, 18]. Consequently, the seemingly contradictory and pleiotropic roles of γδ T cells can be attributed to the intricate complexity of the breast TME. Addressing these nuances is imperative for a comprehensive understanding and effective utilisation of γδ T cells in breast cancer immunotherapy.

Metabolic and functional rewiring of γδ T cells in the TME

Cancer cells are adept at thriving and eluding immune detection through a process known as 'immunoediting' within the TME. The acidic, hypoxic and nutrient-deprived conditions in the TME provide cancer cells with a competitive advantage over immune cells [19, 20]. Studies have revealed that naïve αβ T cells utilise oxidative phosphorylation (OXPHOS) or fatty acid oxidation for ATP production, whereas most effector αβ T cells prefer aerobic glycolysis [21]. In the case of γδ T cells, IFNγ$^+$γδ T cells exhibit a preference for glycolysis, while IL-17$^+$γδ T cells lean towards oxidative metabolism, as evidenced by increased mitochondrial mass and activity [22]. Notably, the hypoxic TME impedes the antitumor effector function of IFNγ$^+$γδ T cells, particularly evident in the context of brain tumours [23]. Similarly, a constrained antitumor effect has been noted in the induction of regulatory CD8$^+$ T cells under hypoxic stimulation [24]. The TME in TNBC has been shown to be markedly hypoxic and immunosuppressive. The elevated hypoxic signature in TNBC patients corresponds to more unfavourable disease prognoses [25]. However, the effects of hypoxic microenvironments on the anti-tumour activity of Vδ2 T cells in TNBC and the impact of Vδ2 T cell allogeneic transfusion in such settings have not yet been documented. The intricate interplay between the TME and the metabolic profiles of immune cells underscores the multifaceted challenges in devising effective immunotherapeutic strategies against cancer.

In this work, we reveal a substantial upregulation of the γδ T cell population in breast tumours compared to normal breast tissues. Utilising The Cancer Genome Atlas (TCGA) database, we establish a positive correlation between γδ T cell abundance and overall survival in breast cancer patients. Our investigation elucidates the functional heterogeneity of γδ T cells at the single-cell level within the contexts of TNBC, non-TNBC, and healthy breast tissues. We unveil cell cycle dysregulation and γδ T cell exhaustion specifically in TNBC tumours. Additionally, we detect metabolic rewiring in γδ T cells infiltrating TNBC, characterised by heightened glycolysis compared to non-TNBC and normal breast tissues. This metabolic shift likely stems from the hypoxic TME of TNBC, modulating transcription factors such as HIF-1α and their corresponding transcriptional networks, thereby influencing downstream effector gene expressions. Consequently, compromised immune responses of γδ T cells in TNBC patients are intricately linked with the hypoxic TME of TNBC tumours. Remarkably, suppression of HIF-1 signalling augments the anti-tumour function of γδ T cell therapy in TNBC tumour-bearing mice. In summary, a comprehensive exploration of the functional roles of γδ T cells in breast cancer, particularly within the TNBC TME, holds significant promise for developing effective adjuvants for allogeneic Vγ9Vδ2 T cell therapy in TNBC treatment.

MATERIALS AND METHODS

Online public databases and bioinformatics analysis

Clinical information and mRNA expression profile data of breast cancer patients and healthy controls were extracted from GEO databases GSE110686, BRCA_GSE150660, GSE161529, GSE176078 and the TCGA database. We analysed tissue samples from the following groups: normal ($n = 112$), triple-negative breast cancer (TNBC, $n = 140$), Luminal A ($n = 419$), Luminal B ($n = 194$) and HER2-enriched ($n = 67$). The MCPCOUNTER and CIBERSORT algorithm were employed to assess the level of γδ T cell infiltration in breast cancer patients, determining the proportion of γδ T cells and the expression of various immune molecules. ChIP-seq data was obtained from the Cistrome Database (http://cistrome.org/db/#/), with specific datasets including IRF1 (GSM1057026), IKZF1 (GSM2827349), RUNX3 (GSM1010893) and STAT1 (GSM1057014).

scRNA-seq analysis

The single-cell transcriptomic data of tumour tissues (GSE110686, GSE161529, GSE176078, BRCA_GSE150660) from both TNBC and non-TNBC patients were grouped using the 10x Genomics ChromiumTM platform. Subsequently, TRDC$^+$CD3$^+$ γδ T cells were extracted from the sequencing results, and the corresponding subgroups of γδ T cells underwent further clustering through dimension reduction to investigate differences in gene expression. Here, we identified 210 γδ T cells from the normal group, 1033 from the TNBC group and 1720 from the non-TNBC group for subsequent analysis. Functional enrichment analysis was then performed. When comparing TNBC versus non-TNBC tumours and TNBC tumours versus normal breast tissues, we employed the GSEABase package (version 1.44.0) to perform gene set variation analysis (GVSA) on gene set files obtained from the KEGG database (https://www.kegg.jp/). Pathway activity of individual cells was assessed using the GSVA package (version 1.30.0), and differences in pathway activity were calculated using the LIMMA package (version 3.38.3). To gain insights into the cell cycle status of the cells, we employed the CellCycleScoring function in Seurat. This involved calculating cell cycle scores for each cell based on the expression of G2/M and S-phase marker genes. These marker sets were selected to be inversely correlated in expression, allowing for the identification of cells in G1 phase that do not express them, as well as those actively cycling in G2/M and S phases. Additionally, the HIF-1 signalling score of γδ T cells in TNBC and non-TNBC was analysed based on the scRNA-seq data using the R package *ggpubr*, following published methods [26].

Breast cancer cell culture

MCF-7, MDA-MB-231, MDA-MB-468, HS578T and SkBr3 cells were obtained from the Cell Bank of Chinese Academy of Sciences and met all quality assurance standards, including STR testing. Cells were cultured in High glucose Dulbecco's modified Eagle's medium (DMEM) (C11995500BT, Gibco) supplemented with 10% foetal bovine serum (F8318, Sigma), 1% Penicillin-Streptomycin (15140122, Gibco) at 37 °C in a humidified atmosphere with 5% CO_2. Cells or their culture supernatant were used for subsequent experiments once cell confluency reached ~80%.

Vγ9Vδ2 T cells expansion

Human peripheral blood was donated by healthy donors. The usage of human peripheral blood was approved by the medical ethics committee of

Jinan University. An informed consent was signed by all subjects. Then peripheral blood mononuclear cells (PBMCs) were isolated from (the) using Ficoll-Paque (17144003, Cytiva) via density gradient centrifugation. The cells were cultured in complete medium Roswell Park Memorial Institute (RPMI) 1640 (C11875500BT, Gibco), supplemented with 10% FBS and 1% Penicillin-Streptomycin. Specifically, Vγ9Vδ2 T cells were activated with 50 nM zoledronate (SML0223, Sigma) in the presence of 400 IU/ml recombinant human interleukin-2 (Beijing Four Rings Bio-Pharm Co). Zoledronate was used to activate the cells only during the first 3 days of culture, after which it was no longer used in the culture medium. In contrast, IL-2 was supplemented in the culture medium throughout the entire cultivation period. Afterward, the cells were harvested for subsequent experiments.

Vγ9Vδ2 T cells cocultured with breast cancer cell lines

To investigate the impact of breast cancer cells on the function of γδ T cells, we first expanded Vγ9Vδ2 T cells derived from donor PBMC in vitro and then cocultured them with MCF-7, MDA-MB-231, MDA-MB-468, HS578T, or SkBr3 breast cancer cell lines under normoxic conditions. The number of cancer cells was 100 million per millilitre, while the number of Vγ9Vδ2 T cells was 500 million per millilitre. After 6 h of coincubation, all cells were harvested using a standardised trypsin digestion protocol. Vγ9Vδ2 T cells were then selectively enriched using CD3 magnetic column separation for further analysis. Parallel-cultured Vγ9Vδ2 T cells without co-culturing with cancer cells were used as the control group. In parallel, for specific experiments, the culture supernatant from these breast cancer cells was collected and directly applied to treat Vγ9Vδ2 T cells to assess the effects of tumour-derived soluble factors. Following these procedures, we performed bulk RNA sequencing (RNA-seq) and metabolic assays to further characterise the functional and molecular changes in γδ T cells.

Analysis of interactions between transcription factors and regulated genes

To ascertain potential common regulatory genes among these transcription factors in γδ T cells, we conducted an examination of ligand-receptor pairs and their corresponding expression levels. This analysis was performed using the CellphoneDB v4.0 software. Subsequently, we constructed a cell interaction network diagram to predict potential communication relationships between various transcription factors and regulatory genes in γδ T cells.

Gene expression level quantification

γδ T cells were cocultured with MCF-7, SkBr3 and MDA-MB-231 cell lines respectively for 6 h, and total RNA was extracted. RNA integrity was evaluated, and then sequenced on the Illumina platform. For the analysis of protein-coding gene expression levels, we used known reference gene sequences and annotation files as a database, and identified the expression abundance of each protein-coding gene in each sample using sequence similarity alignment methods. We used htseq-count software to obtain the number of reads mapped to protein-coding genes in each sample, and calculated the expression level of protein-coding genes FPKM values (https://docs.gdc.cancer.gov/Encyclopedia/pages/FPKM/). For differential expression gene analysis, we used DESeq software to normalise the counts of genes in each sample (using the BaseMean value to estimate the expression level), calculate the differential folds and perform a significance test on the reads using NB (negative binomial distribution test). Finally, we screened differentially expressed protein-coding genes based on the differential folds and significance test results.

Immunoblotting analysis

Total protein was extracted from γδ T cells treated with different conditions using RIPA lysis buffer (P0013B, Beyotime Biotechnology) containing 1% PMSF (HY-B0496, MedChemExpress) and 1% phosphatase inhibitor (B15001, Selleck) and the total protein concentration was subsequently determined using the BCA protein assay kit (P0010S, Beyotime Biotechnology). The same amount of protein was separated by 12% SDS-PAGE gel electrophoresis and transferred to PVDF membrane (IPVH00010, Merck Millipore Ltd.). 5% BSA (V900933, Sigma) dissolved in TBS containing 1% Tween 20 (V900548, Sigma) was used to block non-specific staining sites of samples at room temperature for 1 h, then incubated with the specified primary antibody at 4 °C overnight, followed by washing the membrane with TBST. Then samples were incubated with appropriate HRP-conjugated secondary antibody. The protein bands were observed using an ECL chemiluminescence system (WBKLS0100, Merck Millipore Ltd.) and analysed using ImageJ software. All specific antibodies against were bought from Cell Signalling Technology.

Assessment of cellular metabolism by the Seahorse

Using the Seahorse XFe96 analyser (Agilent), the oxygen consumption rate (OCR) and extracellular acidification rate of γδ T cells treated with 50% MDA-MB-231 supernatant were measured in real time following a standard protocol. This analysis was performed to evaluate mitochondrial function, glycolysis and OXPHOS processes in γδ T cells. In brief, cells were resuspended in the appropriate Seahorse assay buffer and seeded into a pre-coated XF-96 cell culture microplate (Agilent) with 0.1 mg/ml poly-L-lysine (ST508, Beyotime Biotechnology). The plate was then centrifuged to facilitate cell adhesion. For metabolic stress analysis, the following drugs were added: (1) Mitochondrial stress test: 1.5 µM oligomycin, 0.5 µM FCCP (carbonyl cyanide-4-(trifluoromethoxy) phenylhydrazone), and 0.5 µM rotenone/antimycin A; (2) Glycolytic stress test: 10 mM glucose, 1.5 µM oligomycin and 50 mM 2-DG. Finally, key metabolic parameters, including basal respiration, maximal respiration, ATP production, spare respiratory capacity, glycolytic capacity and glycolytic reserve, were calculated.

Immunofluorescence imaging of frozen tissue sections

The tissue sections of breast cancer patients taken at −80 °C were balanced at room temperature for 15 min, then soaked in PBS for 10 min, washed away OCT, closed with PBS containing 5% mouse serum at room temperature for 1 h, and antibodies FITC mouse anti-human CD3 (300306, Biolegend) and BV510 mouse anti-human Vδ2 TCR (743750, Biolegend) were added at 1:50 ratio and incubated at 4 °C for 2 h to dye the membrane surface. Then, we washed with PBS for three times, 5 min each time, added proper amount of BD nucleation fixing solution, at 4 °C for 1 h, drained the fixing solution, rinsed with 1X washing buffer twice, 5 min each time, added PBS containing 5% goat serum and 1% tween 20, and closed at room temperature for 45 min. Then HIF-1α antibody (36169, Cell Signalling Technology) (1:300) was added, incubated at 4 °C overnight, washed three times with PBS for 2 min each time and CF568 (20103, Biotium) labelled goat anti-rabbit fluorescent secondary antibody was added (1: 750), incubated at room temperature for 1 h away from light, washed three times with PBS, finally, added 100 µL anti-quenching agent and observed under a Leica TCS SP8 Laser Scanning Confocal Microscope with a Leica HC PL APO 63x/1.40 Oil CS2 objective after using nail seal tablets.

Atomic force microscopy (AFM) measurements

For information on using AFM to visualise cell surface ultrastructure and measure biomechanical properties, please refer to our previously published work [27, 28]. The biomechanical properties of γδ T cells were measured at room temperature in a liquid cell using AFM (Bioscope Catalyst, Bruker, USA). The spring constant of the cantilever was calibrated to 0.06-0.11 N/m. The comparison of the approach/retraction speed (nm) and distance (nm) curves collected during the entire deflection process was 5 µm/s. The biomechanical properties regarding cell stiffness were calculated from the force curve using Catalyst-equipped software. The average roughness (Ra) of the membrane surface morphology was also generated by the software. The cell height and cell surface area were calculated using the software as well.

Flow cytometry analysis for assessing cell proliferation

To evaluate the proliferation of γδ T cells treated with 50% MDA-MB-231 supernatant, flow cytometry was employed utilising APC anti-human Ki-67 Antibody (350514, Biolegend). Initially, the cell membrane surface was stained with the antibody at room temperature for 15 min, followed by intracellular staining. Subsequently, Ki-67 staining was conducted in accordance with the standard protocol of the eBioscience™ Foxp3/Transcription Factor Staining Buffer Set (00-5523-00, Invitrogen™). The cells were then resuspended in an appropriate amount of PBS before undergoing flow analysis.

Flow cytometry of intracellular and cell suface molecules

All staining steps were performed in dark by following the standard protocol of our lab or of reagent provider. The following antibodies were used to stain the cell surface, including BB515 anti-human TIM-3 antibody (565568, BD Biosciences), BV510 anti-human TCR Vδ2 antibody (331432, biolgend), PE/Dazzle™ 594 anti-human PD-1 antibody (329940, biolgend) and Alexa Fluor® 700 anti-human CD3 antibody (344822, biolgend). For intracellular antigen staining, the single-cell suspension was fixed and permeated using the Foxp3/Transcription Factor Staining Buffer (00-5523-00, Invitrogen™) according to the manufacturer's instructions, followed by staining with PE anti-human HIF-1α antibody (359704, biolgend), or APC anti-human Ki-67 Antibody (350514, Biolegend). For intracellular cytokine staining, cells were performed with 50 ng/mL phorpol-myristate acetate (P8139, Sigma-Aldrich), 1 μg/mL iononomycin (I3909, Sigma-Aldrich) and 2 μM GolgiPlug (555029, BD Biosciences) were incubated at 37 °C for 4 h. Then the cells surface were stained with BV605 anti-human TCR Vδ2 antibody (331430, biolgend) and APC-H7 anti-human CD3 antibody (560176, BD Biosciences). Finally, the cells were fixed and permeabilized using the Fixation/Permeabilization Kit (554714, BD Biosciences), and the following antibodies were used for staining, BV711 anti-human CD107a antibody (328640, biolgend), APC anti-human IFN-γ antibody (554702, BD Biosciences) and PE anti-human TNF-α antibody (502909, biolgend). Prior to antibody staining, cells were blocked using PBS containing 2% mouse serum to avoid nonspecific staining. For cells dissociated from NSI mice γδ T and PX-478+γδ T cell group spleens after grinding and treatment with red cell lysate, were stained according to the intracellular cytokine staining method described above. All samples were resuspended in an appropriate amount of PBS before flow analysis.

NSI mice TNBC model

The immunodeficient NOD-scid-IL2Rg$^{-/-}$ (NSI) mice used in this experiment were purchased from Zhuhai Bestry Biotechnology Co., Ltd. All mice were housed in specific pathogen-free cages and provided with sterilised food and water. The use and protection of animals were approved by the Ethics Committee for Animal Welfare of Jinan University. MDA-MB-231 tumour cells (luciferase-labelled) (1×10^6) were subcutaneously injected into the abdomen of 8 week-old NSI mice, which were then randomly divided into four groups: saline group, γδ T cell group, PX-478 group and PX-478+γδ T cell group, each consisting of eight mice. Female mice were allocated to control or treatment groups without any blinding procedures. The experiment began when the tumour volume reached 70-100 mm^3. At each designated experimental time point, PX-478 (Selleck, 40 mg/kg per mouse, dissolved in a solution containing 4% DMSO and 96% PBS) was intraperitoneally injected at a dose of 40 mg/kg. After 24 h of administration, either 200 μl of 5×10^6 γδ T cells (pre-labelled with DiR dye) suspended in saline or saline alone (control) was intravenously injected into the tumour-bearing mice via tail vein injection. Mice tumour size was measured periodically. The formula for calculating tumour volume is: (width × width × length)/2. The total duration of mouse cell therapy was 23 days, with treatments administered every 3 to 5 days, for a total of 6 treatments. Following ~10 days of observation, two mice in the saline group were found moribund and subsequently died; at this point all remaining animals were humanely euthanized and samples collected. Importantly, tumour burdens in these mice—measured as <20 mm maximum diameter—remained within humane endpoint criteria. The sudden morbidity observed likely reflects the particularly aggressive, highly metastatic nature of the MDA-MB-231 TNBC model. As a result, the measurement on day 34 was the final recorded data point. Before 7 weeks, all mice were sacrificed. To evaluate the functional changes of γδ T cells in the TME of tumour-bearing mice, in vitro expanded γδ T cells were reinfused into the established TNBC (cell line 231) tumour-bearing mice. After 24 h, splenic γδ T cells were isolated for functional analysis. The expression levels of cytotoxic molecules (INFγ, TNFα) of γδ T cells were analysed using flow cytometry. In vivo whole-body imaging of TNBC tumour-bearing mice was carried out using CCD camera system (IVIS 100 Series Imaging System, Xenogen, Alameda, CA, USA).

Statistics

Statistical analysis was performed using GraphPad Prism 9.0 software (GraphPad Software Inc., USA). Statistical significance analysis between multiple groups and between two groups was performed using one-way analysis of variance (ANOVA), Tukey post hoc test, or unpaired Student's *t* test. Data were presented as mean ± standard deviation (SD), unless otherwise specified. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$; ns, no significance.

RESULTS

Tumour-infiltrating γδ T cells positively correlate with better prognosis

To comprehensively delineate the immune landscape within diverse breast cancer TMEs, we conducted an analysis of major functional immune cell subsets utilising data derived from TCGA database. Our investigation centred on the evaluation of infiltration levels pertaining to T cells, natural killer (NK) cells, B cells and monocytes across both normal breast tissue and four distinct breast cancer subtypes, including TNBC, Luminal A, Luminal B, and HER2-enriched (Fig. 1a). The utilisation of MCPCOUNTER scores illuminated significant variances in the infiltration profiles of four different immune cell populations among the different breast cancer subtypes. Intriguingly, conspicuous upregulation was discerned in T, B and monocyte infiltration within TNBC and HER2-enriched subtypes, while remaining negligible in others, suggesting a heterogeneous immune landscape across breast cancer subtypes.

Given the pivotal role of T cell-mediated immunity in orchestrating anti-cancer immune responses, our subsequently focused on γδ T cells, CD4$^+$ T cells and CD8$^+$ T cells (Fig. 1b). Employing CIBERSORT scores, we observed a pronounced augmentation in the tissue-infiltrating of γδ T cells across all four breast tumour types, encompassing TNBC, Luminal A/B and HER2-enriched cancers, while no significant changes were noted in the infiltration levels of CD4$^+$ and CD8$^+$ T cells in TNBC. Crucially, an analysis of overall survival probabilities unveiled that heightened tissue infiltration of γδ T cells (TRDC) correlated with a more favourable prognosis in breast cancer, in contrast to NK (Fig. 1c). Additionally, while CD4$^+$ and CD8$^+$ T cells also showed a positive correlation with better prognosis, their p-values were not as significant as those of γδ T cells (Fig. 1c). Such observations underscore the pivotal role of γδ T cells in conferring anti-tumour functionality among breast cancer patients.

scRNA-seq deciphers transcriptional features of γδ T cell in breast tumours

To unveil the transcriptional features of γδ T cells in breast tumours, we utilised and compiled four GSE datasets (BRCA_GSE150660, GSE176078, GSE161529, GSE110686) from scRNA-seq to thoroughly investigate the infiltration of γδ T cells in breast tumours. Infiltrating γδ T cells were identified by selecting cells with both *CD3* (*CD3D, CD3E, CD3G*) and *TRDC* gene expressions. Initially, we employed Uniform Manifold Approximation and Projection clustering to create 2D graphs comprising 5 clusters, based on γδ T cells within the compiled GSE datasets (Fig. 2a). We then compared the fraction of γδ T cells in these five clusters across three types of tissue, revealing a reduced fraction in clusters 0 and 1, and an increased fraction in clusters 2, 3 and 4 in TNBC tumours (Fig. 2b). The top 10 marker genes for each cluster are presented in Fig. 2c. Further classification of these five clusters of γδ T cells was achieved based on the top expressed marker genes, identifying them as tissue-resident (cluster 0: CD69/IL7R), OXPHOS (cluster 1: CDH1/RHOV), ADCC NK-like (cluster 2: FCGR1A/FCGR2A), proliferating (cluster 3: TYMS/MKI67), and cytotoxic (cluster 4: GZMB/GNLY) (Fig. 2d). Additionally, representative metabolism/signalling pathways for these five clusters, based on gene expression patterns, were depicted in Fig. 2e. Additionally, GSVA analyses were performed to assess the top enriched functional pathways between TNBC and normal tissue, as well as between TNBC and non-TNBC tissue (Fig. 2f). Remarkably, the enrichment of several key metabolic pathways such as glycolysis/gluconeogenesis, pyruvate metabolism and pentose phosphate pathway, etc. in γδ T cells was observed in TNBC compared to non-TNBC and normal tissues.

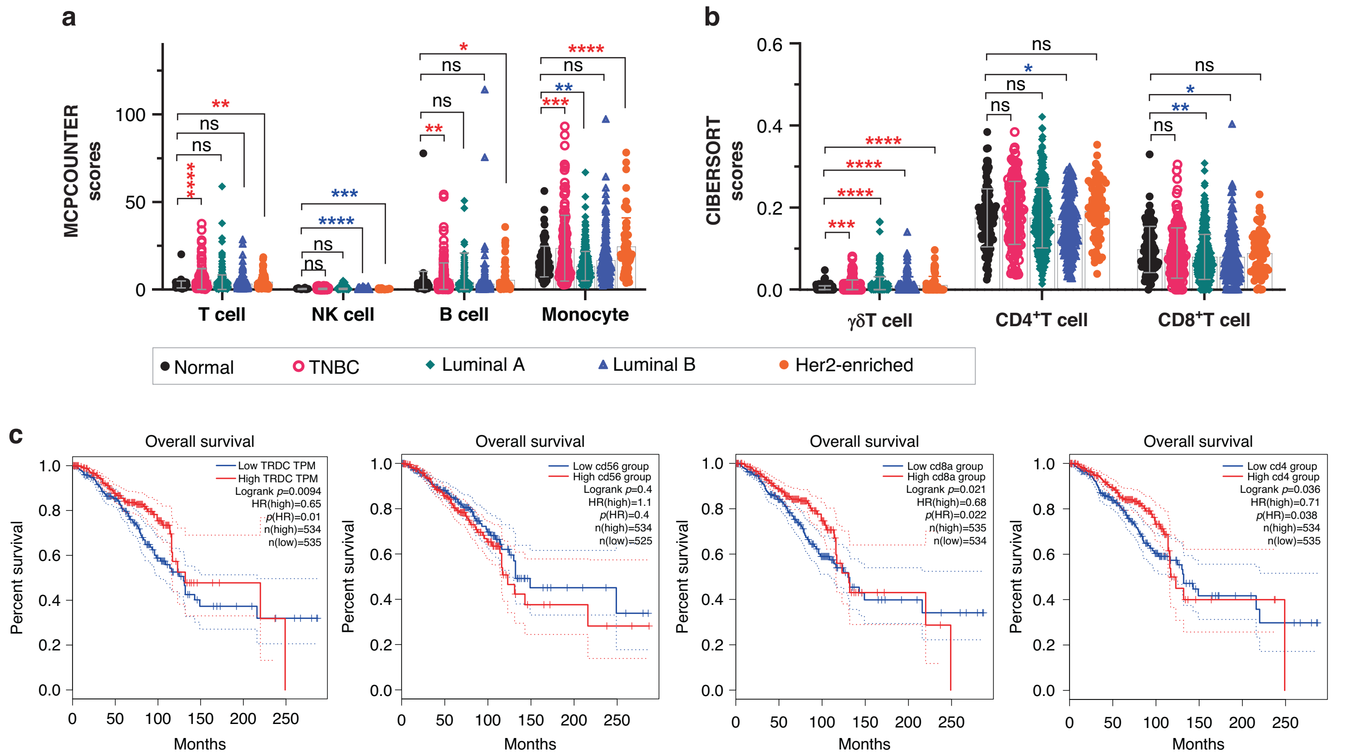


Fig. 1 Bulk-RNA analysis of immune cell infiltration in breast cancer based on TCGA database. a, b Using MCPCOUNTER and CIBERSORT scores, the global infiltration landscape of T cells (CD4$^+$, CD8$^+$ T cell), NK cells, B cells, monocytes and γδ T cells is revealed among normal ($n = 112$), TNBC ($n = 140$), Luminal A ($n = 419$), Luminal B ($n = 194$) and Her2-enriched ($n = 67$) tissue samples. Notably, tissue-infiltrating γδ T cells are remarkably enriched in all four types of breast cancer tissues compared with normal tissues. **c** Overall survival analysis of breast cancer patients associated with γδ T cells (TRDC), NK cells (CD56), CD8$^+$ T cells (CD8A) and CD4$^+$ T cells (CD4), analysed using GEPIA2. It shows tumour-infiltrating γδ T cells predict a better prognosis ($p = 0.0094$). Red *: scores are up in tumour; blue *: scores are down in tumour. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$; ns no significance.

Impaired proliferation and functional exhaustion of TNBC-infiltrating γδ T cells

Given the substantial impact of tissue microenvironment on the anti-tumour activity and proliferation of infiltrating γδ T cells [23, 29, 30], we subsequently conducted KEGG pathway analysis on scRNA-seq data to scrutinise cell cycle-related gene sets across samples. Our analyses unveiled an enrichment of gene sets associated with cell cycle progression and DNA repair mechanisms in γδ T cells within TNBC in contrast to normal or non-TNBC tissues (Fig. 3a), suggestive of active regulation of γδ T cell cycling in TNBC. Further analysis of cell cycle-related gene expression indicates that a higher fraction of γδ T cells in both non-TNBC and TNBC samples appear to be in a non-proliferative state compared to normal tissues (Fig. 3b). This observation, based solely on transcriptomic data, may suggest a transient or sustained withdrawal from the cell cycle. Additionally, γδ T cells in TNBC samples show an obviously reduced fraction expressing markers associated with mitosis (M phase) compared to non-TNBC samples, while the difference between TNBC and normal tissues is less pronounced. Nevertheless, since this analysis is based on gene expression profiles rather than direct protein-level measurements, these findings should be interpreted as preliminary insights that warrant further experimental validation. Additionally, we performed an in vitro experiment in which γδ T cells were cultured with supernatant from the TNBC cell line MDA-MB-231 for 6 h, followed by the analysis of Ki-67 expression via flow cytometry (Fig. S1-a). The data revealed a suppression of γδ T cell proliferation (reduced expression of Ki-67) upon exposure to MDA-MB-231, partially supporting the scRNA-seq analysis depicted in Fig. 3b.

Furthermore, to elucidate the influence of the breast TME on anti-tumour immunity, we conducted a comprehensive analysis of immune effectors and inhibitory checkpoint molecules using bulk RNA-seq data from TCGA. The results revealed a significant upregulation of various effector molecules (such as *IFNG, GZMB*, etc.) and inhibitory checkpoint molecules (including *CTLA4, LAG3, PDCD1*, etc.) at the RNA level across different breast cancer subtypes, particularly pronounced in TNBC tumours (Fig. 3d). To specifically explore functional alterations of γδ T cells from tumours, we extended our investigation of these two panels of makers at the single-cell RNA level as well. This analysis revealed a widespread upregulation of genes linked to effector functions, checkpoint regulation and transcription factors characteristic of IFNγ$^+$ γδ T cells, with this upregulation being particularly pronounced in γδ T cells infiltrating TNBC (Fig. 3e). Figure S1-b illustrates further disparities in the representative expression profiles of cytokines and chemokines in γδ T cells among non-TNBC, TNBC and normal tissues, highlighting substantial heterogeneity within the immuno-landscape of infiltrating γδ T cells. Collectively, these observations suggest that infiltrating γδ T cell in TNBC undergo functional activation (upregulated effector related genes), albeit concomitantly exhibit signs of exhaustion (upregulated checkpoint related genes).

TNBC-infiltrating γδ T cells are metabolically rewired

Considering the immune effector function in γδ T cells is closely regulated by the metabolic processes, we conducted gene set enrichment analysis (GSEA) analysis, particularly focusing on glycolysis and OXPHOS, two fundamental metabolic pathways crucial for effector T cells, utilising scRNA-seq data of infiltrating γδ T cells. The outcomes (Fig. 4a) unveiled that, in comparison to the

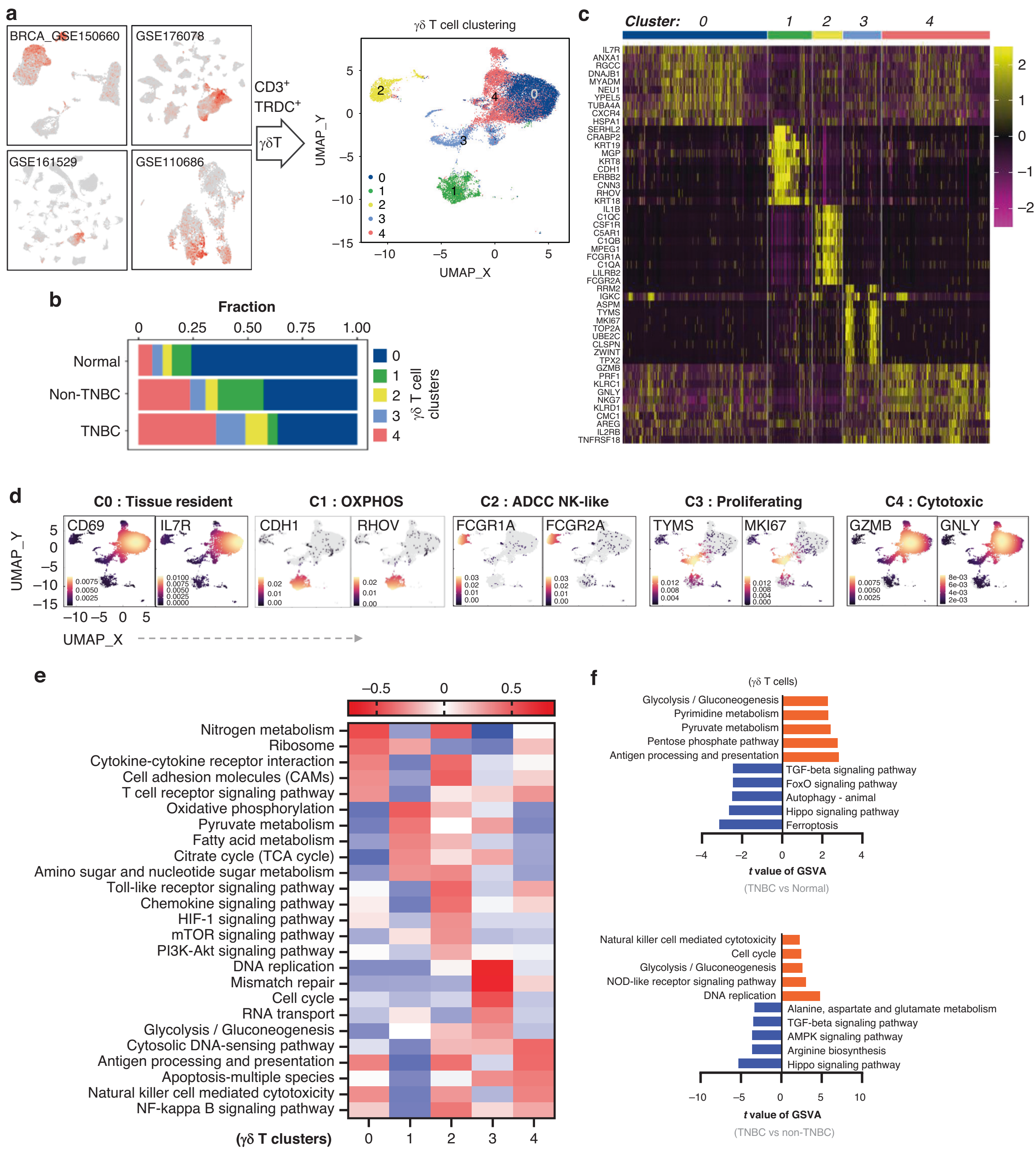


Fig. 2 Single-cell transcriptome of breast cancer tissue-infiltrating γδ T cells. **a** Four single-cell datasets (BRCA_GSE150660, GSE176078, GSE161529, GSE110686) were used to analyze tissue-infiltrating γδ T cells. γδ T cells were identified by selecting cells with both *CD3* (*CD3D, CD3E, CD3G*) and *TRDC* gene expressions. In brief, we identified 210 γδ T cells from the normal group, 1033 from the TNBC group and 1720 from the non-TNBC group for subsequent analysis. **b** Overview of the fractions of each cluster (0, 1, 2, 3, 4) in γδ T cells across three types of tumour samples (normal, TNBC and non-TNBC). **c** Heatmap of marker genes of each cluster. **d** The two most representative marker genes of the five clusters, defining the immune function of each cluster. **e** Enrichment of representative functional pathways for all clusters. **f** GSVA results of top enriched pathways of γδ T cells in TNBC vs normal, or TNBC vs non-TNBC.

normal group, γδ T cells infiltrating TNBC tumours undergo upregulation of both glycolysis and OXPHOS. Furthermore, when contrasted with the non-TNBC group, TNBC-infiltrating γδ T cells exhibit heightened enrichment of glycolysis but not OXPHOS. These data suggest that glycolysis may be associated with modulating antitumor immune responses in TNBC-infiltrating γδ T cells, although this hypothesis requires further experimental validation.

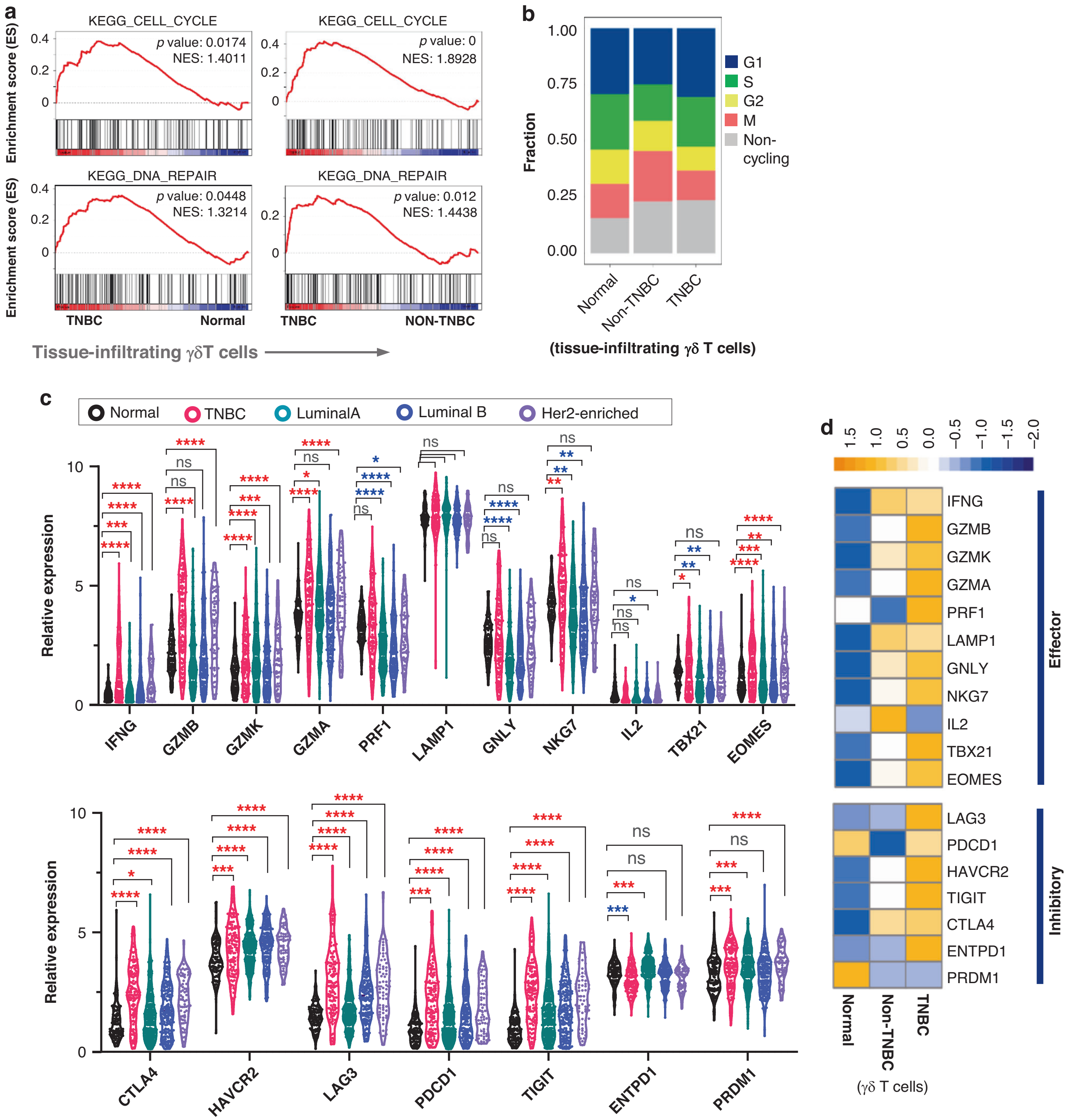


Fig. 3 Functional analyses of single cell RNA-seq and bulk RNA-seq of breast cancer tissue-infiltrating γδ T cells. **a** KEGG pathway enrichment analyses using scRNA-seq data, showing pathways of cell cycle and DNA repair are significantly enriched in TNBC-infiltrating γδ T cells. **b** The fraction of cell cycle phases of γδ T cells across three types of breast tissues, analysed by the R cell cycle package based on expressions of cell cycle related genes. The extended data (In vitro validation of Ki-67 expression in Vδ2$^+$γδ T cells) are shown in Fig. S1-a. **c** Utilising the bulk-RNA data from the TCGA database, the gene expression related to cytotoxic effectors and checkpoints across normal and four different breast cancer tissues was quantitatively compared (normal, $n = 112$; TNBC, $n = 140$; Luminal A, $n = 419$; Luminal B, $n = 194$; Her2-enriched, $n = 67$). Meanwhile, their relative expression values, presented as mean ± SEM (Standard Error of the Mean), are shown in Table S1 (extended data). **d** The scRNA-seq results of γδ T cells visualise the higher expression of genes related to effectors and inhibitory molecules in TNBC, comparing to normal and non-TNBC tissues (extended data are shown in Fig. S1-b). Red *: upregulation in tumour; blue *: downregulation in tumour. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$; ns no significance.

To validate our findings, we used Agilent Seahorse XFe96 Analysers to assess the metabolic activity of in vitro expanded Vδ2$^+$ γδ T cells cocultured with or without MDA-MB-231 cell supernatant. The curves for glycolytic and mitochondrial respiration (OXPHOS) functions are shown in Fig. 4b. Quantitative analysis revealed that while basal respiration and ATP production in γδ T cells were only marginally affected by supernatant treatment, maximal respiration and spare respiratory capacity remained largely unchanged, indicating minimal disruption to OXPHOS. In contrast, glycolytic parameters, including glycolysis rate, glycolytic capacity

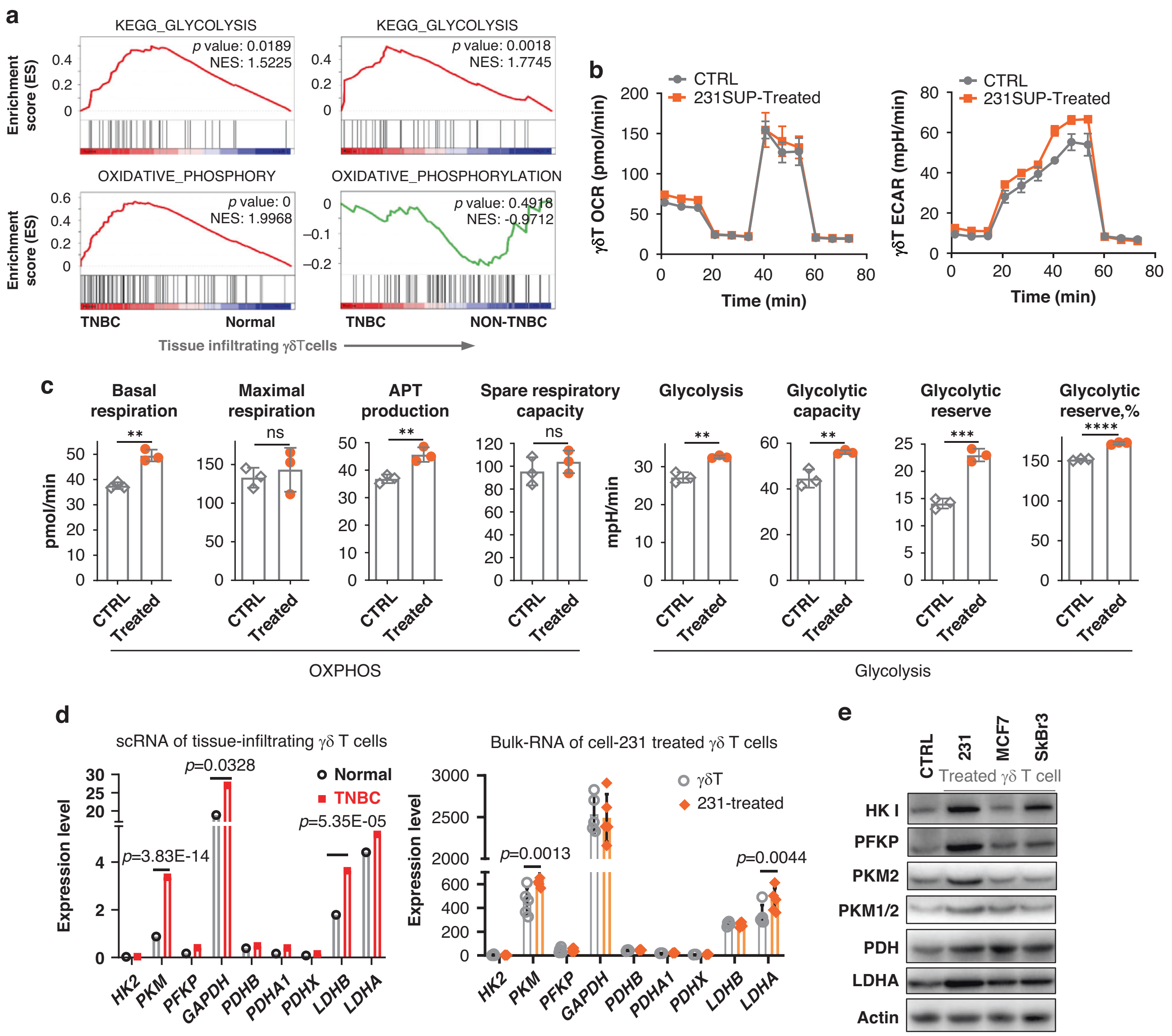


Fig. 4 Metabolic and biophysical characterisation of γδ T cells in the context of breast cancer cell challenge. **a** KEGG enrichment of glycolysis and OXPHOS of infiltrating γδ T cells based on the scRNA-seq data. **b** Mitochondrial respiration, and glycolysis were analysed on MDA-MB-231 supernatant-treated γδ T cells using the Seahorse XF96 Mito Stress and Glycolysis Stress kit, respectively. **c** Statistical results of OXPHOS and glycolysis related metabolic parameters, including basal/maximal respiration, ATP production, spare respiratory capacity, glycolysis and glycolysis capacity/reserve, $n = 3$. **d** The gene expression of key glycolysis enzymes in γδ T cells between TNBC and normal tissue (scRNA-seq data), and between control and cell-231-treated γδ T cells (Bulk RNA-seq data) (extended data are shown in Fig. S2). **e** The WB results for key glycolysis pathway enzymes in γδ T cells treated with MDA-MB-231 (TNBC), MCF7 (Luminal A) and SkBr3 ($HER2^{+}$) supernatants (extended data are shown in Fig. S3). **f**, **g** AFM topography and membrane ultrastructure alterations of γδ T cells with and without cancer cell treatment. **h** Three biophysical parameters, cell height, cell surface area and membrane surface roughness are presented. **i** The schematic diagram of biomechanical acquisition by AFM, and the analysis method of biomechanical properties (cell stiffness and cell membrane surface adhesion force) from force curves. **j** Statistical comparisons of cell membrane surface adhesion force and cell stiffness of γδ T cells in the presence and absence of breast cancer cell treatment, $n > 60$ for each group. **k** KEGG enrichment analysis of the actin cytoskeleton of γδ T cells based on bulk RNA-seq data. Unpaired *t* test was applied for data analyses in (**c**), which represents one of two independent experiments ($n = 3$ per group). *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$; ns no significance.

and glycolytic reserve, were significantly elevated (Fig. 4c), suggesting an enhanced glycolytic metabolism in γδ T cells exposed to the MDA-MB-231 TME. Further RNA analysis revealed that the TNBC TME upregulated *PKM* and *LDH* expression in γδ T cells at both scRNA and bulk RNA levels (Fig. 4d), with weaker effects observed in non-TNBC microenvironments (Fig. S2). Western blot analysis confirmed an obvious upregulation trend of key glycolytic enzymes in γδ T cells cocultured with TNBC cell lines (MDA-MB-231, MDA-MB-468 and HS578T). In contrast, this upregulation trend was less pronounced in γδ T cells cocultured with non-TNBC cell lines (MCF7 and SkBr3) (Fig. 4e and Fig. S3-a). Notably, the changes observed in cell-231 treated γδ T cells can also occur in γδ T cells treated with other two triple-negative breast cancer cells, cell-468 and HS578T (Fig. S3-b). Although minor discrepancies between RNA-seq and protein expression data for key glycolytic enzymes may arise due to post-transcriptional and translational modifications etc., these findings support the conclusion of enhanced glycolysis in γδ T cells exposed to the TNBC TME.

In our previous work, we extensively investigated cellular mechanical changes in response to various stimuli and

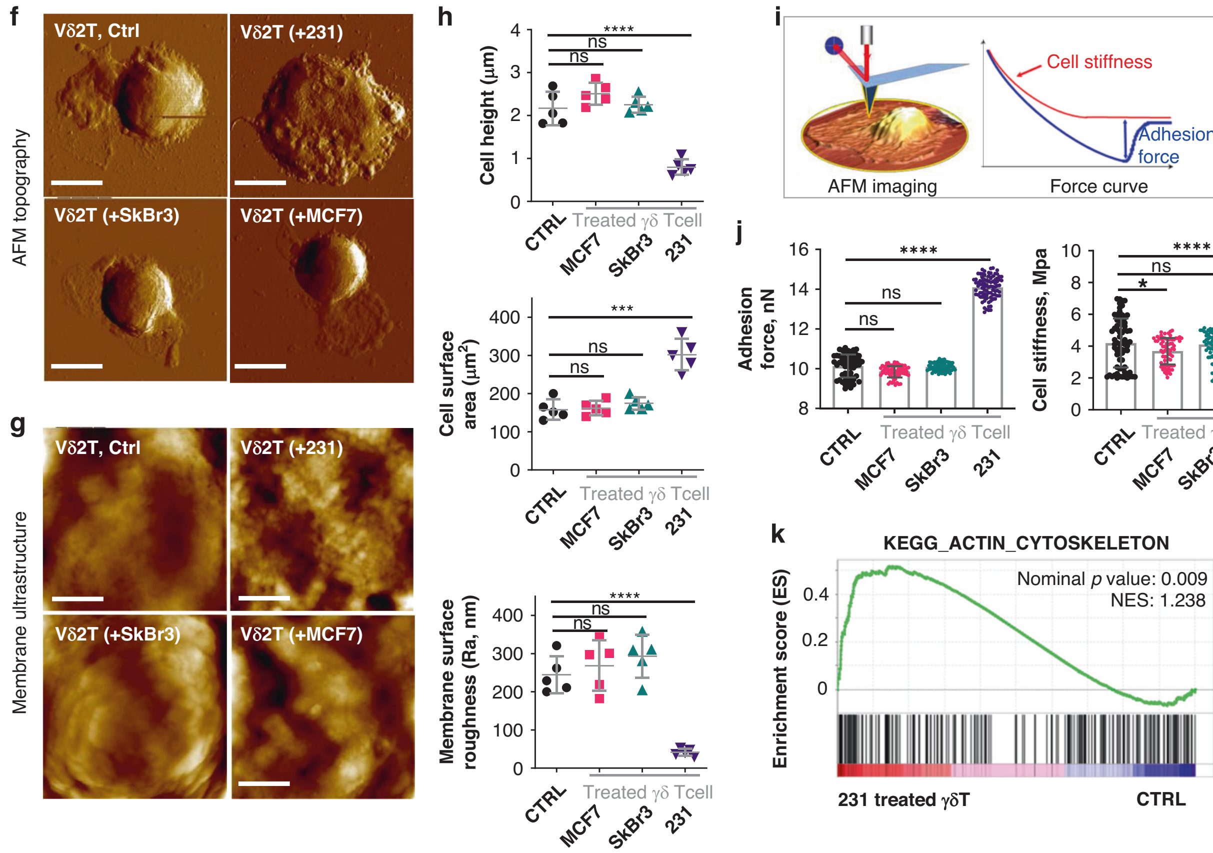


Fig. 4 (Continued)

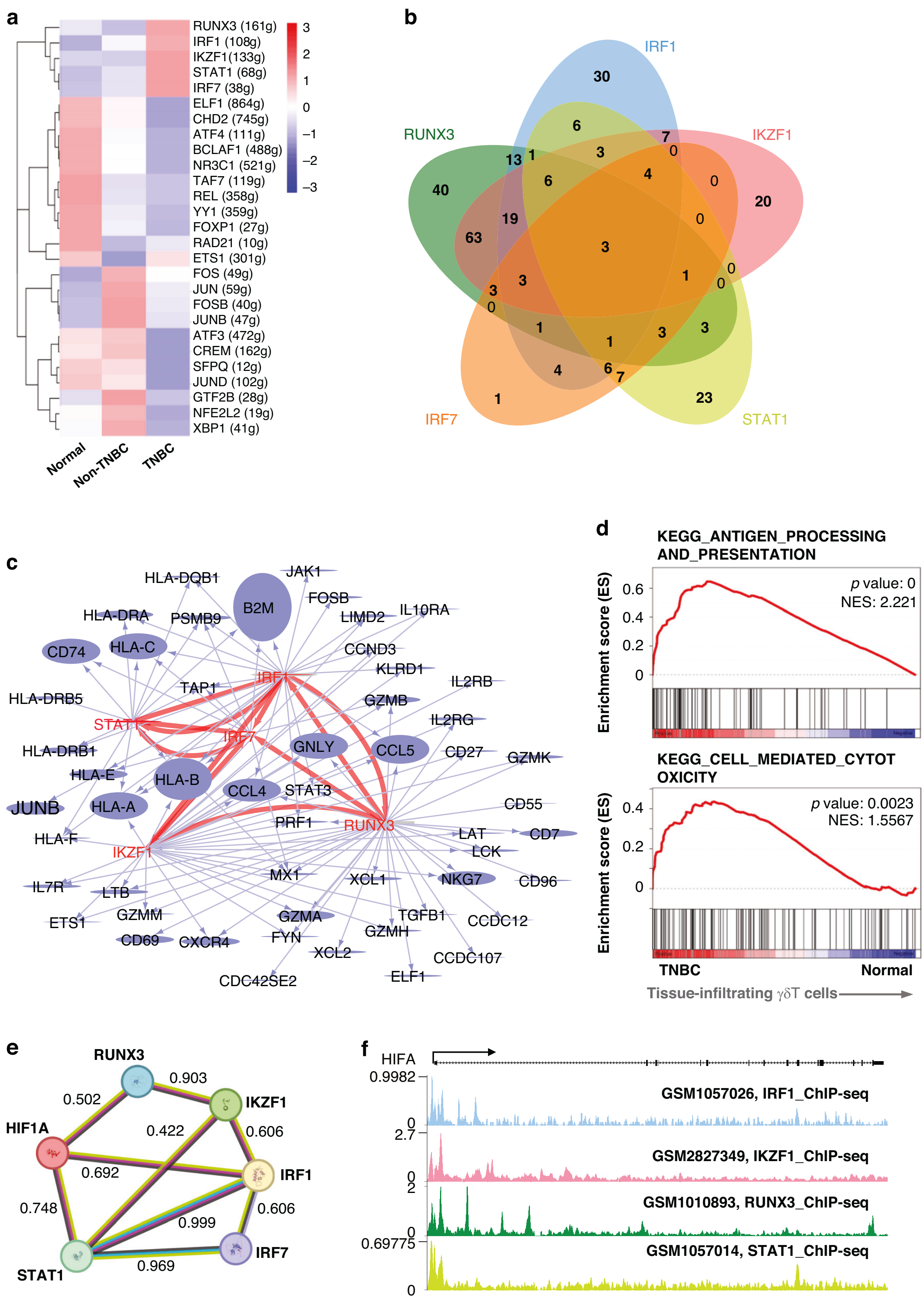
physiological or pathological conditions [27, 28, 31]. For example, we demonstrated that the expression of BRMS1 could modify the ultrastructural, biomechanical and biochemical properties of MDA-MB-435 human breast carcinoma cells [28]. Exploring the biophysics of cells has consistently been a focal point of research endeavours, as biophysical traits are intricately connected to cellular function, particularly energy metabolism.

Given that glycolytic metabolism is pivotal for providing the energy necessary for structural changes, including cytoskeleton reorganisation crucial for immune cell activation, we investigated the impact of MDA-MB-231 cell on the biophysical properties of γδ T cells. This exploration encompassed alterations in topography, ultrastructure and cellular biomechanics. Employing atomic force microscopy (AFM), which can image the cell membrane topography at the nanometre scale and measure biomechanical alterations at the pico-Newton level, we visualised the cell topography (Fig. 4f) and membrane surface ultrastructure (Fig. 4g) of γδ T cells in the absence and presence of breast cancer cell challenge (MDA-MB-231, SkBr3, or MCF7). The topography clearly shows that γδ T cells treated with MDA-MB-231 undergo flattening (Fig. 4f). The ultrastructural images display that, compared to the control, treatment with MDA-MB-231 obviously induced the formation of granules on the cell membrane surface (Fig. 4g). Statistical analyses of cell height, cell surface area and membrane surface roughness were also conducted, revealing that MDA-MB-231 treatment significantly reduced both cell height and membrane roughness (a key parameter for characterising membrane surface ultrastructure) while increasing the cell surface area of γδ T cells (Fig. 4h).

Furthermore, AFM serves not only to visualise cell topography and ultrastructure at the single-cell and nanometre scale but also to quantify biomechanical properties, including cell membrane adhesion force and cell stiffness, at the single cell level. Schematic diagrams in Fig. 4i illustrate how AFM images a cell and how biomechanical properties are derived from the force curves. Common biomechanical properties measured in cells typically include two aspects: cell membrane adhesion force and cell stiffness [28], and both of which are considered as key indicators for T cell activation [32, 33]. Through single-cell force spectroscopy by the AFM, we found that γδ T cells treated with MDA-MB-231 cells significantly induced stronger membrane surface adhesion (left graph in Fig. 4j) and markedly elevated cell stiffness (right graph in Fig. 4j). These findings offer novel evidences of γδ T cell activation upon stimulation with the TNBC cells. Because the level of cell stiffness is predominantly determined by the integrity and organisation of the cytoskeletal actin network [34], we thus conducted bulk RNA-seq of γδ T cells with and without MDA-MB-231 cell stimulation, followed by a KEGG analysis to enrich cytoskeleton-related gene expression. The result unequivocally indicates a significant enrichment of cytoskeletal actin-related genes in the MDA-MB-231-treated γδ T cells (Fig. 4k), corroborating the cell stiffness measurements depicted in the right graph of Fig. 4j at the gene-regulation level. Consequently, our biophysical investigations on breast cancer cell line-stimulated γδ T cells provide new evidence indicating that TNBC-stimulated γδ T cells undergo dynamic alterations in biophysical properties due to activation-induced cytoskeletal reorganisation.

TNBC-infiltrating γδ T cells exhibit a unique pattern of transcription factor regulation

To deepen our comprehension of the transcriptional mechanisms influencing the activated but exhausted phenotypes of TNBC-infiltrating γδ T cells, we analysed the expression of key transcription factors (TFs) in γδ T cells across normal, non-TNBC and TNBC



tumours. TFs are crucial in regulating immune cell effector functions [35, 36]. Notably, five TFs—RUNX3, IRF1, IKZF1, STAT1 and IRF7—were significantly enriched in the TNBC group (Fig. 5a). A Venn diagram illustrated the number of target genes regulated by each TF and their overlaps (Fig. 5b), revealing intricate inter-regulation of genes associated with antitumor effector functions. An interaction network was constructed to visualise these regulatory relationships, highlighting that most target genes are linked to antigen presentation and immune cell-mediated cytotoxicity, supported by KEGG enrichments (Fig. 5c, d). This aligns with observations in Fig. 2c, e, f

Fig. 5 Transcription factor analysis of tissue-infiltrating γδ T cells. a Enrichment heatmap of the most representative transcription factors of γδ T cells among normal, TNBC and non-TNBC tissues, which were analysed based on the scRNA-seq data used in the Fig. 2. **b** Overview of the number of target genes regulated by the five transcription factors (TFs) enriched in TNBC (*RUNX3, IRF7, STAT1, IKZF1, IRF1*). The three genes commonly regulated by the five TFs include *CARD16, USP15 and XAF1*. **c** The regulatory network involving the five TFs and their representative target genes associated with cytotoxicity and antigen presentation. **d** KEGG enrichments highlighting antigen presentation and cytotoxicity in γδ T cells of TNBC compared to normal tissues. **e** Protein-protein interaction between HIF-1α and the five transcription factors RUNX3, IRF7, STAT1, IKZF1, IRF1, analysed using STRING database. The numeric in the graph shows the interaction scores. **f** ChIP-seq data from human leucocytes, sourced from the Cistrome Database, indicate binding peaks of IRF1, IKZF1, RUNX3 and STAT1 around the promoter region of *HIF1A*. Specific datasets include IRF1 (GSM1057026), IKZF1 (GSM2827349), RUNX3 (GSM1010893) and STAT1 (GSM1057014).

and Fig. 3e, indicating an enhancement in the cytotoxic population in TNBC tumour-infiltrating γδ T cells.

Since HIF-1α directly regulates glycolytic gene expression in the TME, and literatures indicate that the hypoxic microenvironment in TNBC impairs anti-tumour responses of tissue-infiltrating immune cells through HIF-1α signalling pathways [25, 37], we explored the role of TFs in regulating HIF-1α (Hypoxia Inducible Factor 1 Subunit Alpha) expression. We used STRING (https://string-db.org/) to analyze protein-protein interactions involving HIF-1α and five key TFs (RUNX3, IRF1, IKZF1, STAT1 and IRF7) that are highly expressed in TNBC tumour-infiltrating γδ T cells. Our analysis identified potential interactions between HIF-1α and RUNX3, IRF1 and STAT1 (Fig. 5e), suggesting a possible regulatory network where HIF-1α coordinates TFs to modulate anti-tumour gene expression (Fig. 5c, d). Further investigation using ChIP-seq data of leucocytes from the Cistrome database revealed that all four TFs bind to the promoter region of the *HIF1A* gene, indicating potential regulatory effects on HIF-1α expression (Fig. 5f).

HIF-1 signalling regulates γδ T cell function and clinical outcomes in TNBC

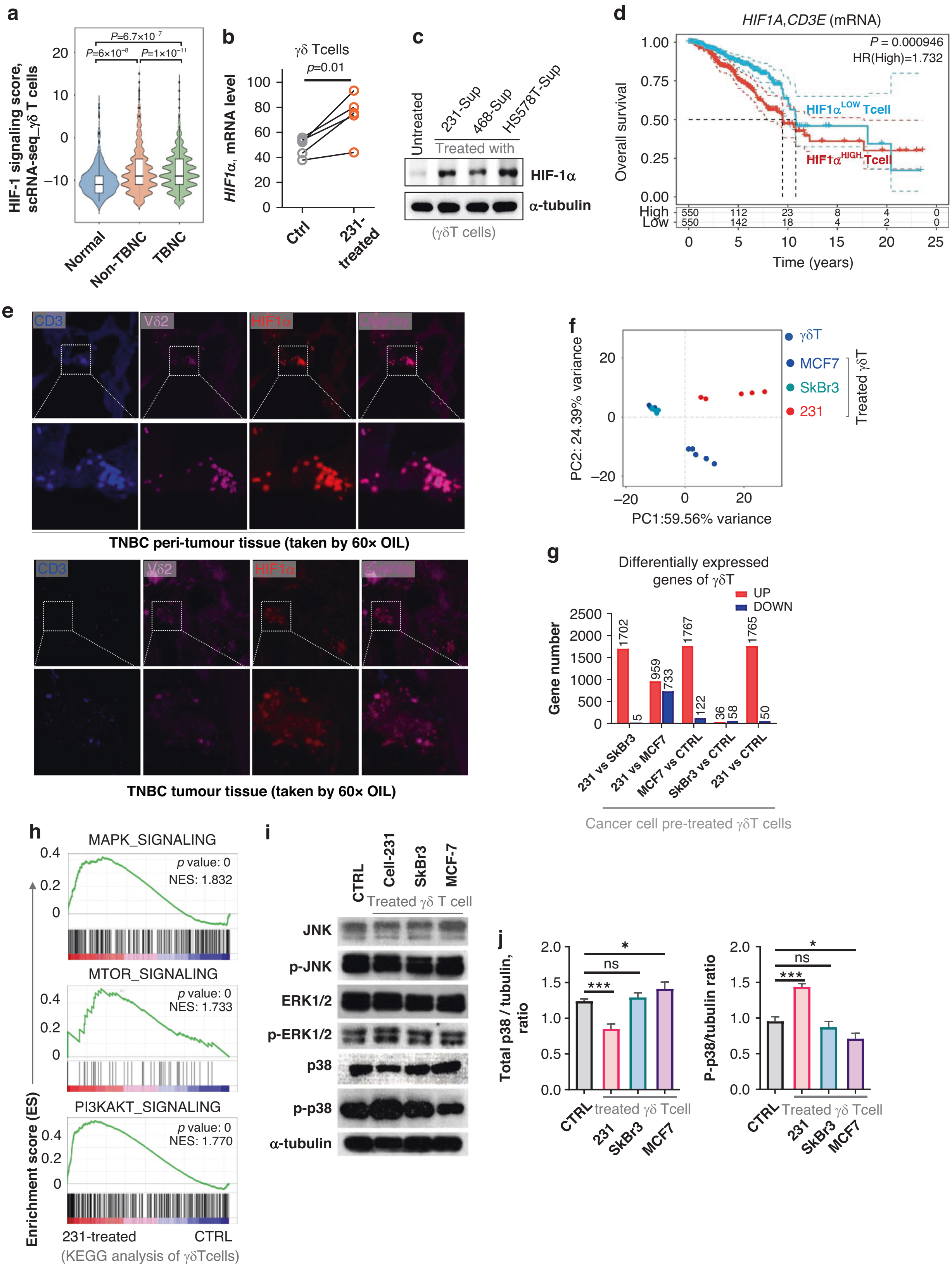
Based on scRNA-seq data of tumour-infiltrating γδ T cells and a previously established HIF-1 signalling score derived from a HIF-1 pathway gene set (VEGFA, SLC2A1, PGAM1, ENO1, LDHA, TPI1, P4HA1, MRPS17, CDKN3, ADM, NDRG1, TUBB6, ALDOA, MIF, ACOT7) [26], we observed that γδ T cells in TNBC exhibit significantly higher HIF-1 signalling scores compared to γδ T cells from normal tissues (Fig. 6a). This result can corroborate with earlier findings indicating heightened HIF-1 signalling in TNBC relative to normal tissues [25, 38]. However, It should be noted that, despite the impressive statistical significance observed between TNBC and non-TNBC, the actual biological difference in HIF-1 signalling scores may be relatively small because the differences in mean scores are modest (Fig. 6a). Given the well-established role of HIF-1α in regulating glycolytic gene expressions under hypoxic conditions [39], we hypothesise that HIF-1α may contribute to the substantial upregulation of glycolysis in γδ T cells in TNBC TME (Fig. 4d, Fig. S2). To further understand how breast cancer cells of different subtypes impact the function of γδ T cells respectively, we further analysed RNA-seq data on γδ T cells cocultured with breast cancer cells under normoxia (20% O_2) condition. It unveiled that TNBC cell line MDA-MB-231 challenge could significantly upregulate *HIF1α* mRNA expression in γδ T cells (Fig. 6b), but not in those co-cultured with SkBr3 cells (Fig. S4). These mRNA level results provided important clues for further exploration of HIF-1α alterations at the protein level in response to the challenge posed by the culture supernatant of breast cancer cells. The WB assay further validated the upregulation of HIF-1α in γδ T cells when challenged with the supernatant of MDA-MB-231 cells (Fig. 6c). This upregulation was also observed when γδ T cells were challenged with the supernatant of two other TNBC cell lines, MDA-MB-468, HS578T (Fig. 6c, Fig. S5-a). These results might be due to that, during their growth, TNBC cells consume significant amounts of ascorbate—a critical cofactor required for PHD (Prolyl Hydroxylase Domain)-mediated HIF-1α degradation. Consequently, the resulting culture supernatant is depleted of ascorbate, which likely leads to the accumulation of HIF-1α in γδ T cells due to reduced degradation. Therefore, we speculate that HIF-1α might be a key regulator responsible for the upregulated glycolysis in γδ T cells exposed to TNBC TME.

To evaluate the clinical relevance of HIF-1α in TNBC, we analysed overall survival probabilities. Our findings indicate that breast cancer patients with low *HIF1α* expression in tumour-infiltrating $CD3^+$ T cells (Fig. 6d) or low enrichment of HIF-1 signalling hallmark gene set (Fig. S6) have a more favourable prognosis. Additionally, confocal fluorescence imaging of TNBC tumour and peri-tumour tissues demonstrated co-expression of γδ T cells with HIF-1α, emphasising the hypoxic milieu encountered by γδ T cells within the TME. Intriguingly, γδ T cells in the peri-TNBC tumour expressed a significantly higher level of HIF-1α compared to those in the TNBC tumour (Fig. 6e). This unexpected finding warrants further investigation.

Given the pivotal role of HIF-1α in coordinating anti-tumour immune responses, we utilised principal component analysis on bulk RNA-seq data obtained from γδ T cells co-cultured with diverse breast cancer cell lines to elucidate genetic alterations in these immune cells (Fig. 6f). The result indicated significant shifts in the gene expression patterns of γδ T cells upon co-culture with both MDA-MB-231 and MCF7 cells, while negligible impact was observed with SkBr3 cells. Further analysis of differentially expressed genes supported this observation (Fig. 6g). We subsequently carried out GSEA analysis focusing on major anti-tumour effector signalling cascades including MAPK, mTOR and PI3K-Akt pathways in γδ T cells treated with cancer cells (Fig. 6h). Results unveiled upregulation of all three pathways in MDA-MB-231-treated γδ T cells, with MAPK pathway exhibiting the most pronounced alteration. This discovery was further confirmed via Western blot analyses, elucidating that exposure to MDA-MB-231 cells triggers activation of the MAPK pathway evidenced by enhanced phosphorylation of p38 and ERK1/2, with p38 phosphorylation demonstrating the most substantial augmentation (Fig. 6i, j). Similar results were (Fig. S5-b) demonstrated using three different types of TNBC cell lines (MDA-MB-231, MDA-MB-468 and HS578T).

Alleviation of HIF-1 signalling restores anti-tumour function γδ T cells in TNBC

To further establish the role of HIF-1 in modulating the anti-tumour immunity of TNBC educated γδ T cells, we first subjected γδ T cell cultures to hypoxic conditions (1% O_2). Results revealed a significant upregulation of HIF-1α in γδ T cells under hypoxic conditions, a response that could be attenuated by PX478, an inhibitor of HIF-1α (Fig. 7a, Fig. S7). Furthermore, effector molecules associated with the anti-tumour function of γδ T cells, such as IFNγ and TNFα, exhibited downregulation under hypoxia but could be restored upon PX478 treatment. Conversely, checkpoint molecules implicated in T cell exhaustion, including PD-1 and TIM3, were upregulated under hypoxia and mitigated by PX478 treatment (Fig. 7a). Additionally, we assessed the phosphorylation levels of two MAPK family members, p38 and ERK1/2 and found that both were elevated under hypoxia conditions. Moreover, phosphorylation of ERK1/2 was notably upregulated to a greater extent. Intriguingly, PX478 treatment only



normalised the phosphorylation level of ERK1/2 but not p38 (Fig. 7b, Fig. S8-a). It should be noted here that, in addition to PX478 used here, another inhibitor, LW6, was also evaluated using WB and flow cytometry. The results indicate that the molecular expression changes observed with PX478 treatment can also be replicated under LW6 treatment (Figs. S8-b, S8-c). This provides solid evidence for further investigation of hypoxic effects on γδ T cell anti-tumour immunity.

Fig. 6 HIF-1α associates with metabolic rewiring of γδ T cells in TNBC. a HIF-1 signalling scores of tissue-infiltrating γδ T cells across normal, non-TNBC and TNBC based on scRNA-seq analysis using the R package *ggpubr*, and the Wilcox test was used for statistical analysis. **b** Bulk RNA-seq validation of *HIF1α* expression of γδ T cells before and after MDA-MB-231 cancer cell line supernatant challenging in vitro (extended data are shown in Fig. S4). Each grey circle represents a control γδ T cell sample from an individual donor, while the corresponding paired sample (after treatment) is shown as orange circle. **c** WB assay of HIF-1α expression of γδ T cells treated without and with the supernatant of TNBC cell lines (MDA-MB-231, MDA-MB-468, HS578T) (extended data are shown in Fig. S5). **d** Correlation between HIF-1 α expression in $CD3^+$ T cells (gene set: HIF1A, CD3E) and overall survival of all types of breast cancers, showing elevated *HIF1A* expression in $CD3^+$ T cells is inversely correlated with patient prognosis (extended data are shown in Fig. S6). **e** HIF-1α expression in tissue-infiltrating γδ T cells in TNBC, acquired through confocal immunofluorescence imaging of frozen tissue sections. **f** Principal Component (PC) visualisation of gene expression pattern of γδ T cells with and without supernatant treatment from cancer cells (MCF7, SkBr3, or 231) using bulk RNA-seq data. **g** Overview of the up- and down-regulated gene numbers in γδ T cells in the absence and presence of cancer cell supernatant treatments, based on bulk RNA-seq data. **h** KEGG enrichments of three representative metabolism regulation pathways, MAPK, MTOR and PI3K/AKT respectively. **i** WB validation of JNK, ERK and p38 expression of γδ T cells before and after cancer cell treatment. **j** Statistical analysis of total p38 and phosphorylated p38 (P-p38). $*p < 0.05$; $**p < 0.01$; $***p < 0.001$; $****p < 0.0001$; ns no significance.

Subsequently, we conducted in vivo mouse experiments to assess whether suppressing HIF-1 signalling could restore the anti-tumour function of γδ T cells. To evaluate the clinical applicability of HIF-1α inhibitors, we utilised PX478, a HIF-1α inhibitor that has shown promise in phase I clinical trials [40], in combination with adoptive transfer of human $V\delta2^+$γδ T cells to treat subcutaneous TNBC tumours in NSI mice, which lack effective T, B and most NK cell populations (Fig. 7c). Colocalization of fluorescence signals from inoculated tumours and adoptively transferred $V\delta2^+$γδ T cells was evident in both the γδ T cell-treated group and the group treated with γδ T cells combined with PX478 (Fig. 7d). Results demonstrated significantly slower tumour growth in the group receiving γδ T cells plus PX478 administration compared to the group receiving γδ T cells alone, indicating inhibition of HIF-1 signalling enhanced the anti-tumour immunity of γδ T cells (Fig. 7e). Additionally, we analysed the expressions of IFNγ and TNFα of $V\delta2^+$γδ T cells before transferring into mice and after exposure to the TNBC TME (Fig. 7f). The results revealed that in vitro expanded $V\delta2^+$γδ T cells expressed IFNγ and TNFα at rates of 88% and 85.4%, respectively (Fig. 7g, Fig. S9). However, both IFNγ and TNFα expression were suppressed in $V\delta2^+$γδ T cells after in vivo exposure to the TNBC TME for 24 h. It should be noted here that co-administration of PX478 with γδ T cell transfer partially restored the expression of IFNγ and TNFα in γδ T cells, suggesting a partial restoration of anti-tumour function (Fig. 7g). In conclusion, our findings highlight the critical role of HIF-1 signalling in modulating the anti-tumour function of γδ T cells in TNBC. By demonstrating the efficacy of PX478, a HIF-1α inhibitor, in enhancing the therapeutic potential of γδ T cell-based treatment, our study provides valuable insights into the clinical management of TNBC. These results underscore the importance of targeting the hypoxic TME to improve the effectiveness of immunotherapeutic approaches in breast cancer treatment.

DISCUSSION

Recent advances have leveraged allogeneic γδ T cells for the treatment of advanced-stage cancers, including liver (NCT03183219), lung (NCT03183232) and breast cancers (NCT03183206) [10, 11]. While tumour-infiltrating γδ T cells are identified as promising prognostic indicators and play a crucial role in anti-tumour immunity [9, 29], our clinical trial data reveal variability in clinical outcomes. This variability highlights the urgent need to thoroughly understand the mechanisms regulating γδ T cell functions within the TME.

Breast cancer, characterised by its complexity and heterogeneity, presents diverse microenvironments that significantly impact disease progression and treatment outcomes. Particularly, TNBC is known for its aggressive nature and poorer prognosis compared to other subtypes. Our study, which explores γδ T cells in the context of TNBC, offers novel insights into potential targeted intervention strategies. A comprehensive analysis of immune cell subsets within breast cancer TMEs, with a focus on γδ T cells due to their emerging importance in anti-tumour immunity, reveals intricate dynamics that could influence prognosis and therapeutic approaches.

Analysis of TCGA database revealed substantial variations in immune cell infiltration across different breast cancer subtypes. TNBC and HER2-enriched subtypes exhibited notable increases in T, B and monocyte cell infiltration compared to other subtypes, indicating a heterogeneous immune landscape. Notably, our study identified significant augmentation of γδ T cell infiltration across all breast cancer types. A key finding was the positive correlation between γδ T cell infiltration and overall survival in breast cancer patients, suggesting that heightened γδ T cell presence is associated with improved prognosis. This correlation underscores the potential of γδ T cells as both prognostic biomarkers and therapeutic targets in breast cancer.

To investigate the role of γδ T cells in breast tumours, we utilised scRNA-seq to explore their transcriptional features and functional states. Our analysis revealed distinct γδ T cell subpopulations with varying functions, including tissue-resident, proliferating, cytotoxic and metabolically active subsets. This functional diversity underscores the importance of considering these subtypes in therapeutic strategies. In particular, γδ T cells within TNBC tumours displayed signs of impaired proliferation and functional exhaustion, evidenced by dysregulated cell cycle progression and increased expression of inhibitory checkpoint molecules. Metabolic profiling further indicated heightened glycolysis in these TNBC-infiltrating γδ T cells, suggesting that they undergo metabolic reprogramming in response to the TME. This increased glycolytic activity is associated with γδ T cell activation and structural changes, emphasising the impact of the TME on their function.

In-depth analysis uncovered a distinct pattern of transcription factor regulation in γδ T cells within TNBC tumours, which may explain the observed metabolic and functional alterations. TFs such as *RUNX3*, *IRF1* and *STAT1* were significantly enriched, indicating their potential roles in modulating the expression of effector and checkpoint molecules. ChIP-seq data suggested that these TFs might regulate *HIF1A* gene expression, a key regulator of glycolytic genes. This regulatory mechanism could account for the elevated glycolysis observed in γδ T cells. These findings provide crucial insights into the complex transcriptional regulation of γδ T cell metabolism and anti-tumour functions in breast cancer, highlighting potential targets for therapeutic intervention.

Our study also identified HIF-1 as a critical regulator of γδ T cell function in TNBC. Hypoxic microenvironments have been reported to inhibit the antitumor function of γδ T cells infiltrating brain tumours [23]. However, the impact of hypoxic microenvironments on γδ T cells infiltrating TNBC and the antitumor effects of allogeneic γδ T cell transfusion have not yet been reported. Here, we found that HIF-1α signalling is involved in the upregulation of glycolysis and activation of the MAPK pathway. Moreover, we

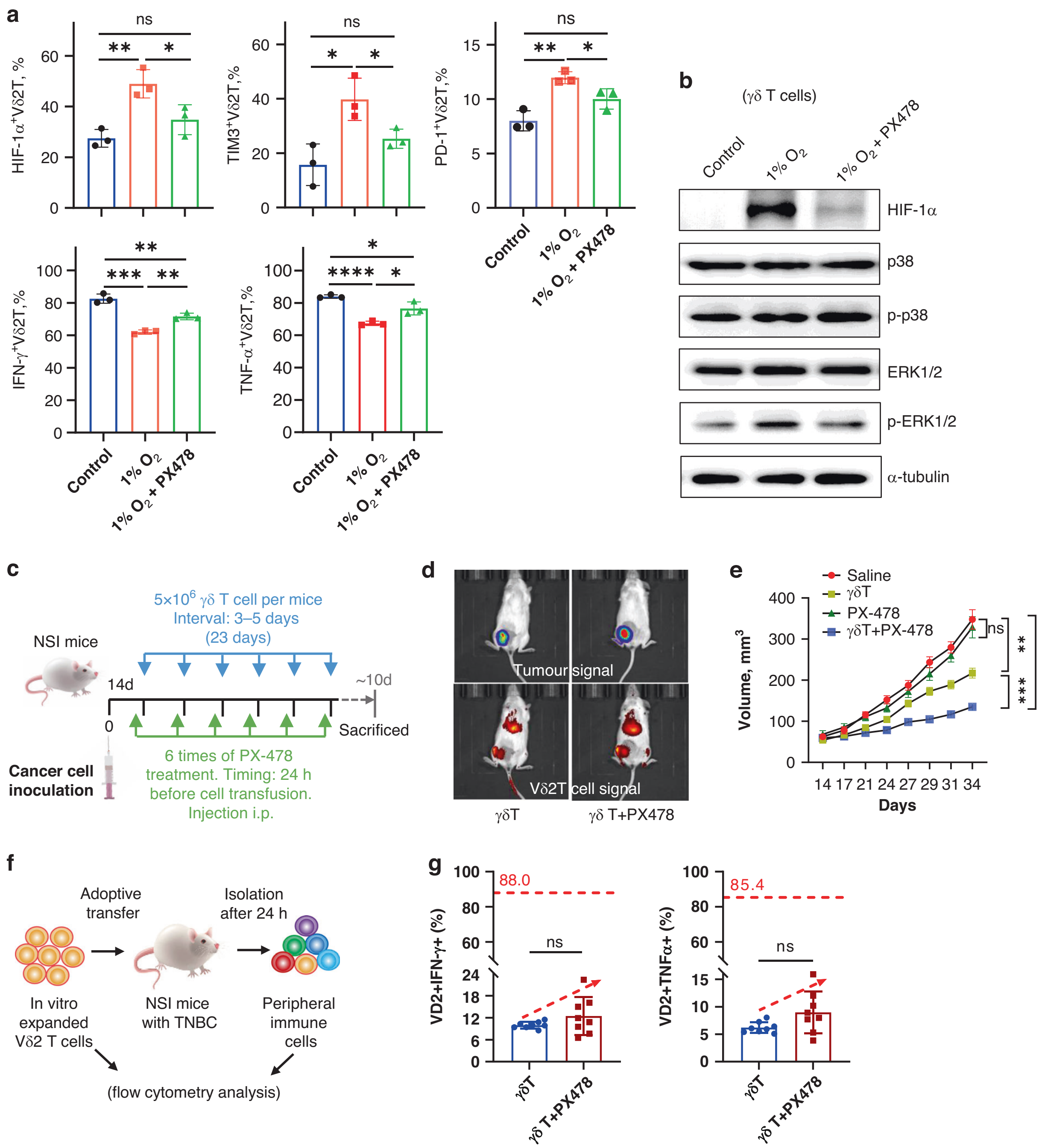


Fig. 7 In vitro and in vivo validation of HIF-1α in regulating the cytotoxicity of γδ T cells. **a** Expression of HIF-1α, TIM3, PD1, IFNγ and TNFα in Vδ2$^+$γδ T cells in the context of normoxia (20% O_2, Control group), hypoxic condition (1% O_2) and the hypoxic inhibitor (PX478) (extended data are shown in Fig. S7), analysed by flow cytometry. **b** WB validation of expression of p38 and ERK1/2, as well as their phosphorylation status, in Vδ2$^+$γδ T cells treated under normoxia, hypoxia condition (1%) alone and hypoxia condition (1%) together with PX478 ($n = 3$ per group) (extended data are shown in Fig. S8-a). In addition to PX478, another inhibitor, LW6, was also evaluated using WB (Fig. S8-b) and flow cytometry (Fig. S8-c). **c** Schematic diagram of mouse model, displaying processes of tumour inoculation, cell therapy, inhibitor application and sample collection. **d** Representative images from two groups acquired by the in vivo imaging system. MDA-MB-231 cells were fluorescence transfected (luciferase) prior to inoculation, and Vδ2$^+$γδ T cells were stained with the dye DiR prior to adoptive transfer into mice. **e** Tumour volume of control groups (Saline, PX478) and therapy groups (γδ T, γδ T + PX478), data were presented as mean ± SEM ($n = 8$). **f** A schematic diagram illustrating Vδ2$^+$γδ T cells before their transfer into mice and subsequent isolation from the spleen (24 h of exposure) for analysis via flow cytometry. This approach aims to assess alterations in the functionality of Vδ2$^+$γδ T cells induced by the tumour-bearing microenvironment in mice. **g** Expression of IFNγ and TNFα in Vδ2$^+$γδ T cells isolated from the tumour-bearing mice 24 h after adoptive transferred, showing that adjuvant usage of PX478 shows a trend toward counteracting the significant suppression of these two effectors induced by the tumour microenvironment (paired *t*-test used here). The red dashed line and numeric in two graphs shows the expression levels of these two molecules in Vδ2$^+$γδ T cells before transferring into mice. $n = 8$ mice per group. Extended data are shown in Fig. S9. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$; ****$p < 0.0001$; ns no significance.

demonstrated that targeting HIF-1α could enhance γδ T cell-mediated anti-tumour immunity. In preclinical models, pharmacological inhibition of HIF-1α, combined with γδ T cell adoptive transfer, resulted in synergistic tumour growth suppression. These results suggest that the alleviation of the hypoxic TME could significantly improve the efficacy of γδ T cell immunotherapy in breast cancer.

In conclusion, our study provides valuable insights into the immune landscape of breast cancer, with a focus on γδ T cells. By elucidating the transcriptional, functional and metabolic characteristics of γδ T cells within breast tumours, we have identified potential targets for immunotherapy and proposed strategies to overcome immune evasion in TNBC. Despite these insights, our research has limitations, including reliance on bioinformatics analyses from databases like TCGA and GEO, potential batch effects in single-cell RNA-seq data and the representativeness of in vitro experiments. Further studies are needed to validate the role of HIF-1 in γδ T cell function and to explore advanced techniques such as tissue mass spectrometry and single-cell proteomics. It should be particularly noted that our cell-based experimental model can only partially simulate the effects of tumour cells on γδ T cells within the TME, and cannot fully replicate the physiological changes in vivo. This represents another limitation of our current study. In future similar studies, we plan to utilise mouse models to address such issues. Addressing these limitations will enhance our understanding and therapeutic strategies for breast cancer.

DATA AVAILABILITY

Data are provided within the manuscript or supplementary information files.

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AUTHOR CONTRIBUTIONS
Work supervision and design: YH and YZW. Experiments: YH, YYJ, YZW and QLH. Clinical samples and data collection: YH, YYJ, XH, DL, WHH and MGF. Data analysis and discussion: YH, YZW and YYJ. Mouse model: YYJ, QLH and WFW. Manuscript writing and revision: YH, YZW and YYJ. All authors approved the final version of manuscript.

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COMPETING INTERESTS
The authors declare no competing interests.

ETHICS APPROVAL
The animal related experiment was approved by the Jinan University Laboratory Animal Ethics Committee (#20210712-28).

CONSENT FOR PUBLICATION
All co-authors hereby grant their consent for the publication of our work, understanding that it may be shared publicly.

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Correspondence and requests for materials should be addressed to Yangzhe Wu, Minggang Fu, Xin Huang or Yi Hu.